

STUDENT PERFORMANCE ANALYSIS
PROJECT REPORT

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Project Summary:

This Project focuses Prediction on Student Performance Analysis using python, then by managing the student database through queries and Analysing the performance based on the features of the dataset through Attractive Visualization plots. To predict How effective students performed based on their academic behavior, study habits, lifestyle routines, and exam conditions.

Dataset:

I have used preprocessed Dataset - [Exam Score Prediction Dataset](#), with 500 Observations. So there is no need for preprocessing techniques such as Handling Missing Values, Null Values or Outliers.

Columns:

- Age,
- Gender,
- Course,
- Study method,
- Study hours,
- Sleep hours,
- Sleep Quality,
- Internet access,
- Facility rating,
- Exam Difficulty,
- Exam Score.

Software & Techniques:

SQL – Managing the student Database and handling them as per requirements using Queries.

Python – Predicting the student performance based on the course and exam score.

R – Visualizing the Analysis.

Tools:

- My SQL Workbench
- Jupyter Notebook
- R Studio

SQL

Queries:

- Created a New Schema name “student”.
- Just for understanding Checking the null values for Each column.
- Analysing the count of students in each course.

	course	count(*)
▶	diploma	72
	bca	88
	b.sc	64
	b.tech	75
	bba	73
	ba	53
	b.com	75

First Analyzing the Top Scorers:

- Getting the data where the student scores above > 90.

course	exam_score
b.sc	100
bba	100
ba	100
b.sc	100
b.tech	100
ba	100
bca	100
bca	100
b.com	100
diploma	100
b.tech	100
bba	100
b.com	100
b.com	100
diploma	99.3
bba	98.5
bba	97.9
b.tech	97.5
bba	97.5
bba	97.1
ba	96.4
b.sc	96.3
b.tech	96.1
bba	96
b.com	95.6
diploma	95.5
b.sc	95.2
ba	93.8
bca	93.8
diploma	93.1
ba	92.2
b.tech	92.1
b.tech	92
ba	91.6
b.tech	91.3
bca	90.9
ba	90.7
bba	90.6
ba	90.6
b.sc	90.3

- Count of students scored exactly 100 in their courses and exam difficulty.

course	exam_sco	exam_diffi	count(*)	
b.sc	100	easy	1	
ba	100	moderate	1	
b.sc	100	moderate	1	
b.tech	100	moderate	1	
ba	100	easy	1	
bca	100	hard	1	
diploma	100	moderate	1	
b.tech	100	hard	1	
b.com	100	moderate	2	
bca	100	moderate	1	
bba	100	moderate	2	
b.com	100	hard	1	

- Count of students scored from 95- 99 in their courses and exam difficulty.

course	exam_sco	exam_diffi	count(*)	
bba	98.5	moderate	1	
b.tech	97.5	easy	1	
b.sc	95.2	moderate	1	
diploma	95.5	easy	1	
b.com	95.6	moderate	1	
b.tech	96.1	easy	1	
bba	97.5	moderate	1	
ba	96.4	easy	1	
bba	96	hard	1	
b.sc	96.3	moderate	1	
bba	97.9	moderate	1	
bba	97.1	moderate	1	

Analyzing the top scorers academic behaviour and lifestyle routines.

- 100 Scorers

age	gender	course	exam_diffi	study_hou	sleep_hou	sleep_qual	study_method	
24	male	b.sc	easy	7.83	7.8	good	coaching	
20	other	ba	moderate	7.72	9.9	good	group study	
22	male	b.sc	moderate	7.91	8.8	poor	mixed	
20	female	b.tech	moderate	6.65	6.8	average	mixed	
17	female	ba	easy	6.11	9.3	good	group study	
22	female	bca	hard	7.35	6.6	average	coaching	
22	female	diploma	moderate	7.68	7.8	average	coaching	
20	male	b.tech	hard	7.29	9.6	good	coaching	
24	male	b.com	moderate	7.86	7.9	good	coaching	
22	female	bca	moderate	6.43	7.8	poor	online videos	
17	female	bba	moderate	7.39	6.8	average	mixed	
24	female	bba	moderate	5.61	9.8	average	self-study	
23	female	b.com	moderate	6.11	9	average	coaching	
22	female	b.com	hard	7.85	9.2	average	mixed	

- Between 95 - 99 scorers

age	gender	course	exam_diffi	study_hou	sleep_hou	sleep_qual	study_met	exam_score
23	other	bba	moderate	4.84	7.3	average	coaching	98.5
17	male	bba	moderate	5.8	5.5	good	coaching	97.9
20	other	b.tech	easy	7.28	6.3	poor	mixed	97.5
22	male	bba	moderate	7.22	4.9	good	online vide	97.5
21	female	bba	moderate	7.84	5.3	poor	group stud	97.1
19	male	ba	easy	6.96	4.4	average	coaching	96.4
21	male	b.sc	moderate	7.8	9.5	good	mixed	96.3
19	female	b.tech	easy	7.11	5.9	poor	mixed	96.1
18	other	bba	hard	5.22	6.5	good	mixed	96
22	male	b.com	moderate	6.26	5.1	good	coaching	95.6
22	male	diploma	easy	7.77	9	average	coaching	95.5
20	other	b.sc	moderate	5.7	7.3	poor	coaching	95.2

Analyzing the Average Scorers (50-75) their Academic Behaviour and Lifestyle Routines

- Attaching the sample data from 70 average scorers count due to Large number of data.

course	exam_score	exam_diff	count(*)
bba	74.9	moderate	1
b.com	74.8	moderate	1
b.com	74.8	hard	1
bca	74.8	easy	1
b.tech	74.5	moderate	1
diploma	74.4	moderate	1
bca	74.3	easy	1
ba	74.3	moderate	1
b.sc	74.2	hard	1
ba	74.1	hard	1
b.tech	74	moderate	1
diploma	73.9	moderate	1
bca	73.7	moderate	1
b.sc	73.7	easy	1
b.com	73.6	moderate	1
b.com	73.6	hard	1
b.sc	73.1	easy	1
b.tech	73	easy	1
ba	72.9	easy	1
bba	72.7	moderate	1

- Attaching the sample data from 60 scorers their life style routine

17	male	b.com	hard	6.9	9.5	average	coaching	70.1
21	other	ba	easy	3.97	9.5	good	group stuc	69.9
19	other	bba	moderate	5.23	5.2	poor	mixed	69.9
18	female	diploma	easy	4.81	5.7	good	mixed	69.9
19	other	diploma	hard	7.76	6.3	poor	online vide	69.8
22	female	bca	moderate	6.05	4.3	average	group stuc	69.7
18	male	b.com	moderate	5.74	4.2	poor	online vide	69.6
17	other	diploma	moderate	3.8	8.3	poor	online vide	69.6
20	male	bba	hard	3.73	8.3	good	mixed	69.5
21	other	b.sc	moderate	7.12	4.1	poor	mixed	69.5
19	male	bba	easy	0.74	9.4	average	mixed	69.5
20	other	diploma	hard	7.69	7.4	average	self-study	69.4
22	other	b.com	easy	6.64	9.5	poor	online vide	69.4
19	female	b.sc	moderate	5.96	5.5	average	self-study	69.1
23	female	bba	easy	4.11	5.3	good	group stuc	69.1
23	male	bba	moderate	0.91	8	average	coaching	68.9
19	other	b.sc	hard	3.24	9.4	average	self-study	68.8
20	male	ba	moderate	7.63	8.5	good	online vide	68.6
23	male	b.sc	moderate	3.9	7.1	average	mixed	68.4
18	male	bca	easy	3.4	6.7	poor	online vide	68.4
21	female	ba	easy	5.23	4.2	poor	coaching	68.1
23	female	bca	moderate	5.5	5.1	poor	coaching	68.1
19	male	b.tech	moderate	5.7	9.2	poor	self-study	67.9
19	female	b.tech	easy	3.67	8.3	good	mixed	67.9
19	other	diploma	easy	6.45	4.4	poor	group stuc	67.9
18	male	b.sc	easy	2.38	7.5	poor	mixed	67.7
18	other	diploma	moderate	5.89	9.5	average	mixed	67.6
21	female	bca	easy	0.92	5.5	good	online vide	67.6
21	other	diploma	easy	2.98	6.2	good	group stuc	67.6
22	female	ba	moderate	5.74	5.8	poor	self-study	67.4
19	male	b.sc	moderate	7.69	9.9	poor	group stuc	67.2
17	male	diploma	easy	1.24	7.7	average	mixed	67.1
24	female	bca	hard	5.46	9.9	good	mixed	67

Let's Analyze the difference between Top Scorers and Average Scorers Class attendance, utilizing Internet Access, Facility rating

- 100 Scorers

age	gender	class_atten	internet_a	facility_rat	exam_difficulty
24	male	67.8	yes	high	easy
20	other	85.3	yes	medium	moderate
22	male	62.7	yes	high	moderate
20	female	96.4	yes	low	moderate
17	female	85.9	no	high	easy
22	female	96.2	yes	medium	hard
22	female	82.7	yes	low	moderate
20	male	69.5	yes	high	hard
24	male	65.9	yes	high	moderate
22	female	85.8	yes	high	moderate
17	female	94.8	yes	high	moderate
24	female	93.2	yes	high	moderate
23	female	89.9	yes	high	moderate
22	female	63.9	yes	low	hard

- **95-99 Scorers**

age	gender	class_atten	internet_a	facility_rat	exam_diffi	exam_score
23	other	99.4	yes	high	moderate	98.5
17	male	88.7	yes	high	moderate	97.9
20	other	83.9	yes	high	easy	97.5
22	male	84.6	yes	high	moderate	97.5
21	female	89.8	yes	low	moderate	97.1
19	male	83.3	yes	medium	easy	96.4
21	male	90.3	yes	medium	moderate	96.3
19	female	94.5	no	medium	easy	96.1
18	other	77.2	no	high	hard	96
22	male	49.6	yes	high	moderate	95.6
22	male	91.2	yes	medium	easy	95.5
20	other	92.1	yes	high	moderate	95.2

- **50-75 Scorers – Sample of 50 Exam Score Scorers**

19	other	65.8	yes	high	moderate	59.1
19	male	72.1	yes	high	moderate	59
17	male	92.9	yes	low	hard	58.9
19	female	67.3	yes	high	easy	58.9
21	male	54.9	yes	low	hard	58.3
23	male	65.4	yes	low	moderate	58.2
20	female	54.8	yes	medium	easy	58.2
19	female	99.4	yes	high	moderate	57.9
18	other	46.6	yes	medium	easy	57.8
18	female	80.7	yes	high	easy	57.7
22	male	99.4	yes	low	easy	57.6
18	female	59.4	yes	low	moderate	57.6
20	other	40.6	yes	low	easy	57.5
18	male	49.9	no	low	hard	57.5
22	female	57.4	yes	low	moderate	57.4
24	male	44	yes	medium	moderate	57.3
17	male	90.7	yes	low	moderate	57.3
24	female	46.1	yes	medium	moderate	57.2
24	female	92.6	yes	high	moderate	57.2
19	other	80.2	yes	high	easy	57.2
18	other	61.6	yes	medium	easy	57
20	female	90.1	no	low	moderate	57
19	other	53	yes	medium	easy	57

Let's Analyze the Poor Scorers – Below 50 Scorers

- Sample Count of Below 50 Scorers

course	exam_sco	exam_diffi	count(*)
bca	19.599	moderate	1
ba	19.599	moderate	1
b.tech	19.599	easy	1
diploma	19.599	easy	1
b.sc	19.599	easy	1
bca	19.7	moderate	1
b.tech	19.8	easy	1
b.tech	21.4	easy	1
b.tech	22.1	hard	1
b.tech	22.9	hard	1
bba	23.5	easy	1
b.tech	23.7	easy	1
diploma	23.9	moderate	1
ba	25.5	moderate	1
bba	26.4	easy	1
diploma	26.5	moderate	1
diploma	26.7	moderate	1
b.sc	27	easy	1
b.tech	27.2	hard	1
b.tech	27.6	moderate	1
bba	27.7	easy	1
b.tech	28	easy	1
b.com	28.3	moderate	1
bca	28.3	moderate	1

- Their Academic Behaviour and Life Style Routines

age	gender	course	exam_diffi	study_hou	sleep_hou	sleep_qual	study_met	exam_score
19	other	bca	moderate	0.08	9.6	poor	mixed	19.599
17	male	ba	moderate	2.28	5.5	poor	self-study	19.599
19	male	b.tech	easy	0.19	5.5	poor	online vide	19.599
18	male	diploma	easy	0.08	6.3	poor	coaching	19.599
17	male	b.sc	easy	1.09	4.6	average	mixed	19.599
24	other	bca	moderate	1.29	5.4	poor	self-study	19.7
18	male	b.tech	easy	1.1	5.8	poor	group stud	19.8
22	female	b.tech	easy	0.94	5.9	good	self-study	21.4
20	female	b.tech	hard	0.13	6.1	poor	online vide	22.1
17	male	b.tech	hard	1.76	7.9	poor	group stud	22.9
23	male	bba	easy	2.53	9.4	average	self-study	23.5
18	female	b.tech	easy	0.19	7	poor	group stud	23.7
22	female	diploma	moderate	0.43	5.9	average	self-study	23.9
18	female	ba	moderate	0.39	5.8	poor	coaching	25.5
23	female	bba	easy	0.08	5.5	good	coaching	26.4
21	female	diploma	moderate	0.08	5.4	poor	coaching	26.5
19	male	diploma	moderate	1.33	5.7	poor	online vide	26.7
22	female	b.sc	easy	1.73	7.7	good	mixed	27
18	male	b.tech	hard	2.48	4.8	poor	online vide	27.2
17	male	b.tech	moderate	0.74	4.6	good	group stud	27.6
17	other	bba	easy	0.15	5.8	average	self-study	27.7
24	other	b.tech	easy	0.11	6	poor	group stud	28
22	female	b.com	moderate	1.35	5.6	poor	coaching	28.3
23	male	bca	moderate	2.26	4.1	average	online vide	28.3

- Sample data of Their Facility rating and utilizing Internet Access and Class Attendance

age	gender	class_atten	internet_a	facility_rat	exam_diffi	exam_score
19	other	50.9	yes	low	moderate	19.599
17	male	65.6	yes	medium	moderate	19.599
19	male	44	yes	low	easy	19.599
18	male	63.4	no	low	easy	19.599
17	male	42.1	yes	low	easy	19.599
24	other	56	yes	high	moderate	19.7
18	male	57.4	no	medium	easy	19.8
22	female	61.1	yes	medium	easy	21.4
20	female	82.7	yes	high	hard	22.1
17	male	58.2	yes	low	hard	22.9
23	male	41.6	no	medium	easy	23.5
18	female	48.9	yes	medium	easy	23.7
22	female	62.6	yes	low	moderate	23.9
18	female	87.3	yes	low	moderate	25.5
23	female	42.2	no	high	easy	26.4
21	female	94.8	yes	high	moderate	26.5
19	male	82.4	yes	high	moderate	26.7
22	female	54.9	yes	low	easy	27
18	male	47.3	yes	low	hard	27.2
17	male	77	yes	low	moderate	27.6
17	other	75.5	yes	low	easy	27.7
24	other	71.4	yes	medium	easy	28
22	female	43.6	yes	medium	moderate	28.3
23	male	51.4	yes	low	moderate	28.3

Let's Analyze the student performance of Top Scorers, Average Scorers and Poor Scorers based on the gender

- Male 100 Scorers

	age	gender	course	study_hours	class_attendance	study_method	exam_difficulty
▶	24	male	b.sc	7.83	67.8	coaching	easy
	22	male	b.sc	7.91	62.7	mixed	moderate
	20	male	b.tech	7.29	69.5	coaching	hard
	24	male	b.com	7.86	65.9	coaching	moderate

- Female 100 Scorers

age	gender	course	study_hou	class_atten	study_met	exam_difficulty
20	female	b.tech	6.65	96.4	mixed	moderate
17	female	ba	6.11	85.9	group stud	easy
22	female	bca	7.35	96.2	coaching	hard
22	female	diploma	7.68	82.7	coaching	moderate
22	female	bca	6.43	85.8	online vide	moderate
17	female	bba	7.39	94.8	mixed	moderate
24	female	bba	5.61	93.2	self-study	moderate
23	female	b.com	6.11	89.9	coaching	moderate
22	female	b.com	7.85	63.9	mixed	hard

- Other gender 100 Scorer

	age	gender	course	study_hours	class_attendance	study_method	exam_difficulty
►	20	other	ba	7.72	85.3	group study	moderate

- Male 95 -99 scorers

	age	gender	course	study_hours	class_attendance	study_method	exam_difficulty	exam_score
►	17	male	bba	5.8	88.7	coaching	moderate	97.9
	22	male	bba	7.22	84.6	online videos	moderate	97.5
	19	male	ba	6.96	83.3	coaching	easy	96.4
	21	male	b.sc	7.8	90.3	mixed	moderate	96.3
	22	male	b.com	6.26	49.6	coaching	moderate	95.6
	22	male	diploma	7.77	91.2	coaching	easy	95.5

- Female 95-99 Scorers

	age	gender	course	study_hours	class_attendance	study_method	exam_difficulty	exam_score
►	21	female	bba	7.84	89.8	group study	moderate	97.1
	19	female	b.tech	7.11	94.5	mixed	easy	96.1

- Other Gender 95-99 Scorers

	age	gender	course	study_hours	class_attendance	study_method	exam_difficulty	exam_score
►	23	other	bba	4.84	99.4	coaching	moderate	98.5
	20	other	b.tech	7.28	83.9	mixed	easy	97.5
	18	other	bba	5.22	77.2	mixed	hard	96
	20	other	b.sc	5.7	92.1	coaching	moderate	95.2

- Male Average Scorers sample

age	gender	course	study_hou	class_atte	study_met	exam_diffi	exam_score
24	male	bba	1.84	86.7	coaching	moderate	74.9
23	male	bca	5.01	55.7	mixed	easy	74.8
22	male	b.com	5.92	93.6	self-study	moderate	74.8
20	male	diploma	4.72	97.5	self-study	moderate	74.4
23	male	ba	5.03	98.5	mixed	hard	74.1
19	male	diploma	5.96	70.6	mixed	moderate	73.9
21	male	b.sc	7.91	50	group stud	easy	73.7
19	male	b.com	5.07	59.7	group stud	moderate	73.6
22	male	bba	6.38	77.2	mixed	moderate	72.7
21	male	b.com	7.54	50.1	group stud	moderate	72.6
21	male	bca	3.02	91.6	group stud	moderate	72.5
21	male	ba	6.26	50.3	self-study	hard	72.4
20	male	b.tech	5.42	65.7	mixed	moderate	72.2
20	male	bca	5.76	55.7	self-study	easy	71.6
17	male	bca	4.32	58.5	coaching	easy	71.4
23	male	b.tech	6.9	87.1	online vide	hard	71.1
18	male	bba	6.2	92.5	online vide	moderate	71
17	male	b.tech	6.26	56.9	self-study	easy	70.9
18	male	ba	4.41	90.8	group stud	hard	70.6
22	male	b.com	2.55	96.1	online vide	easy	70.6
18	male	diploma	5.44	94.9	coaching	easy	70.5
21	male	b.sc	1.97	91.9	self-study	moderate	70.3
21	male	bca	5.73	55.1	group stud	easy	70.2
17	male	b.com	6.9	78.1	coaching	hard	70.1

- Female Average Scorers sample

18	female	diploma	4.81	51.2	mixed	easy	69.9
22	female	bca	6.05	64.8	group stud	moderate	69.7
23	female	bba	4.11	70.7	group stud	easy	69.1
19	female	b.sc	5.96	83.5	self-study	moderate	69.1
23	female	bca	5.5	72	coaching	moderate	68.1
21	female	ba	5.23	60.4	coaching	easy	68.1
19	female	b.tech	3.67	71.2	mixed	easy	67.9
21	female	bca	0.92	96.5	online vide	easy	67.6
22	female	ba	5.74	96.6	self-study	moderate	67.4
23	female	bba	4.11	49.2	group stud	easy	67
20	female	diploma	4.37	78.7	online vide	easy	67
24	female	bca	5.46	48.8	mixed	hard	67
19	female	diploma	5.23	55.5	coaching	moderate	66.8
19	female	bba	4.72	62.3	coaching	easy	66.7
19	female	b.sc	6.68	44.6	group stud	moderate	66.3
23	female	bba	2.72	47.4	online vide	hard	66.1
19	female	b.tech	5.23	54.5	online vide	easy	66.1
22	female	b.tech	1.82	62.7	self-study	moderate	65.9
21	female	bca	4.9	87.8	group stud	easy	65.9
20	female	bca	3.68	73.1	online vide	easy	65.8
23	female	bba	5.5	59.3	self-study	hard	65.8
24	female	bca	3.66	60.8	online vide	easy	65.4
19	female	b.tech	5.04	69.8	self-study	easy	65.3
20	female	bca	1.91	97.2	self-study	easy	64.7
23	female	diploma	4.13	72.7	mixed	moderate	64.6
19	female	b.com	3.98	53.8	mixed	easy	64

- Other gender Average Scorers Sample

17	other	bca	4.05	44.5	online vide	easy	60.2
22	other	bba	5.83	53.2	coaching	moderate	59.6
17	other	bba	1.4	86.3	mixed	moderate	59.4
19	other	b.tech	3.91	65.8	mixed	moderate	59.1
18	other	bca	3.23	46.6	self-study	easy	57.8
20	other	bca	2.83	40.6	mixed	easy	57.5
19	other	bba	2.97	80.2	online vide	easy	57.2
18	other	bba	1.23	61.6	mixed	easy	57
19	other	bca	4.7	53	mixed	easy	57
21	other	b.tech	0.53	95.8	mixed	easy	56.9
20	other	bca	0.21	49.2	coaching	moderate	56.4
24	other	diploma	3.78	45.9	self-study	hard	56.1
20	other	b.sc	2.34	69.7	group stud	moderate	55.4
23	other	bca	3.37	64.8	online vide	moderate	54.8
24	other	b.com	2.66	80.5	mixed	moderate	54
22	other	diploma	3.92	44.7	mixed	moderate	54
21	other	bca	1.36	75.5	self-study	easy	53.3
24	other	b.tech	2.32	82.6	self-study	moderate	53.1
19	other	bca	4.07	59.3	online vide	moderate	52.4
17	other	bca	3.88	67.7	coaching	hard	52.3
22	other	diploma	3.26	42.6	coaching	moderate	52.2
19	other	ba	1.7	70.4	mixed	easy	52.1
21	other	b.tech	1.58	82.1	group stud	hard	51.9
23	other	diploma	4.95	42	online vide	moderate	51.5
23	other	bca	1.46	93.7	mixed	hard	51.1
19	other	bca	0.68	87.2	self-study	moderate	50.6

- Male Poor Scorers

age	gender	course	study_hc	class_att	study_mi	exam_di	exam_score
17	male	ba	2.28	65.6	self-stud	moderate	19.599
19	male	b.tech	0.19	44	online vic	easy	19.599
18	male	diploma	0.08	63.4	coaching	easy	19.599
17	male	b.sc	1.09	42.1	mixed	easy	19.599
18	male	b.tech	1.1	57.4	group stu	easy	19.8
17	male	b.tech	1.76	58.2	group stu	hard	22.9
23	male	bba	2.53	41.6	self-stud	easy	23.5
19	male	diploma	1.33	82.4	online vic	moderate	26.7
18	male	b.tech	2.48	47.3	online vic	hard	27.2
17	male	b.tech	0.74	77	group stu	moderate	27.6
23	male	bca	2.26	51.4	online vic	moderate	28.3
21	male	bca	2.12	53	self-stud	moderate	30.3
22	male	b.sc	0.25	78.6	online vic	easy	31.9
24	male	bca	1.08	95	group stu	moderate	32
24	male	bca	2.18	68	online vic	moderate	32.1
18	male	bca	0.4	64.9	self-stud	moderate	32.3
19	male	bca	1.89	52	group stu	moderate	35.5
24	male	b.com	0.27	91.7	group stu	moderate	36.3
18	male	b.tech	0.08	74	group stu	easy	36.6
22	male	bca	0.79	55.4	coaching	moderate	37.6
20	male	bba	2.5	40.6	group stu	moderate	37.8
19	male	b.com	4.97	47.7	mixed	hard	37.9
18	male	bba	1.1	50.4	online vic	hard	38.8
24	male	b.com	0.19	44.5	coaching	hard	38.9
22	male	b.com	3.68	56.4	group stu	easy	39.7
22	male	diploma	0.99	57.4	online vic	moderate	40
19	male	b.tech	4.62	56.5	online vic	moderate	40.2
19	male	b.sc	0.45	65.5	group stu	moderate	42.9
22	male	b.sc	2.66	75.9	self-stud	hard	44.3
24	male	b.com	1.61	86.1	online vic	hard	44.7
17	male	bca	4.03	64.7	self-stud	moderate	45.7
21	male	b.com	0.49	85.1	mixed	moderate	46.2
18	male	ba	1.44	86.8	group stu	easy	47.1
17	male	b.sc	2.08	63.9	mixed	easy	47.2
22	male	b.sc	5.48	51.1	self-stud	moderate	47.3
18	male	b.com	3.41	58.1	self-stud	easy	47.4
19	male	b.tech	0.91	60	group stu	moderate	47.4
22	male	ba	3.13	88.6	online vic	moderate	47.4
17	male	ba	3.36	64.6	mixed	moderate	49
24	male	diploma	3.71	79.2	online vic	easy	49
18	male	diploma	0.17	65.4	self-stud	moderate	49.1
20	male	b.tech	4.11	59.9	coaching	moderate	49.3
18	male	ba	2.73	47.2	mixed	moderate	49.7

- Female Poor Scorers

age	gender	course	study_hc	class_att	study_mi	exam_di	exam_score
22	female	b.tech	0.94	61.1	self-stuc	easy	21.4
20	female	b.tech	0.13	82.7	online vi	hard	22.1
18	female	b.tech	0.19	48.9	group stu	easy	23.7
22	female	diploma	0.43	62.6	self-stuc	moderate	23.9
18	female	ba	0.39	87.3	coaching	moderate	25.5
23	female	bba	0.08	42.2	coaching	easy	26.4
21	female	diploma	0.08	94.8	coaching	moderate	26.5
22	female	b.sc	1.73	54.9	mixed	easy	27
22	female	b.com	1.35	43.6	coaching	moderate	28.3
22	female	b.sc	0.9	74.5	mixed	moderate	30.9
24	female	b.tech	0.61	46	group stu	moderate	31.4
24	female	b.com	2.85	48.4	mixed	moderate	32.1
23	female	bba	0.81	66	online vi	moderate	34.1
23	female	bba	4.26	49.4	group stu	easy	34.3
21	female	b.com	1.61	48.4	self-stuc	moderate	34.3
24	female	b.tech	1.24	43.1	online vi	easy	34.8
17	female	bba	1.21	45.6	online vi	moderate	35.1
24	female	b.sc	2.37	47	group stu	moderate	35.4
21	female	bba	1.28	59.1	self-stuc	moderate	35.6
21	female	bba	1.91	40.6	coaching	easy	36.7
19	female	bca	2.17	79.6	self-stuc	hard	37
23	female	b.sc	0.12	43.2	online vi	hard	37.3
24	female	bba	1.79	71.6	self-stuc	moderate	37.3
20	female	b.sc	1.17	62.7	mixed	moderate	39
23	female	b.com	2.22	42.8	online vi	moderate	39.8
23	female	bca	3.44	55.8	mixed	moderate	40.8
19	female	ba	3.88	72.5	self-stuc	easy	41.8
18	female	b.sc	2.2	79.1	coaching	moderate	42.5
17	female	b.com	1.72	77.2	group stu	easy	43
20	female	diploma	0.89	71.6	coaching	moderate	43.7
20	female	b.tech	0.47	86.6	online vi	moderate	45
21	female	bba	4.27	52.7	self-stuc	moderate	45.7
21	female	diploma	1.09	53.6	self-stuc	moderate	45.7
20	female	b.com	1.17	75.2	online vi	easy	46.9
22	female	bca	5.42	70.2	coaching	moderate	47.8
17	female	diploma	1.35	99.4	group stu	moderate	48.1
20	female	b.sc	2.84	63.7	group stu	moderate	48.5

- Other Gender Poor Scorer

age	gender	course	study_h	class_at	study_m	exam_di	exam_scor
19	other	bca	0.08	50.9	mixed	moderat	19.599
24	other	bca	1.29	56	self-stuc	moderat	19.7
17	other	bba	0.15	75.5	self-stuc	easy	27.7
24	other	b.tech	0.11	71.4	group st	easy	28
23	other	diploma	1.61	44.9	mixed	easy	29.2
20	other	diploma	0.67	48.4	online vi	moderat	29.7
23	other	b.tech	0.7	63.5	mixed	moderat	31.3
22	other	diploma	1.23	50.1	coachini	moderat	31.4
24	other	ba	2.42	41.7	online vi	moderat	31.7
19	other	b.sc	1.61	76	group st	easy	32
22	other	b.com	0.6	55	online vi	easy	33.6
18	other	b.com	0.81	61.2	self-stuc	hard	37.1
23	other	bca	2.46	70.4	mixed	hard	37.1
20	other	bca	0.57	74.1	self-stuc	moderat	38.1
22	other	bca	0.32	47.8	self-stuc	hard	38.1
17	other	diploma	1.5	49.7	group st	easy	38.3
19	other	bca	2.1	68.3	online vi	hard	38.8
17	other	diploma	2.64	58.1	self-stuc	moderat	39.5
24	other	diploma	3.47	81.2	group st	moderat	39.6
20	other	ba	1.75	86.7	online vi	moderat	39.9
21	other	bba	2.45	76.1	self-stuc	easy	40.7
20	other	diploma	0.08	67.7	online vi	easy	40.8
21	other	diploma	1.6	54	coachini	hard	41
17	other	b.com	1.88	52.8	mixed	hard	41.1
20	other	b.tech	0.4	57.2	mixed	easy	42.3
17	other	b.sc	0.75	72	group st	moderat	42.8
18	other	b.sc	0.95	49.2	group st	hard	43.2
24	other	bca	1.51	59.3	mixed	easy	44.3
22	other	bca	2.96	89.2	self-stuc	easy	44.4
18	other	bca	2.89	92	self-stuc	easy	44.9
18	other	diploma	3.76	51.4	self-stuc	moderat	45
18	other	ba	0.19	72	self-stuc	moderat	45.3
24	other	b.sc	0.9	44.9	self-stuc	moderat	45.4
17	other	b.tech	1.54	52.5	mixed	moderat	45.5
20	other	diploma	3.36	55.9	self-stuc	easy	45.6
18	other	b.sc	0.28	40.9	self-stuc	easy	46
21	other	b.com	2.54	46.7	coachini	moderat	48.3
22	other	diploma	2.69	93.6	coachini	moderat	49
21	other	bca	2.96	48.5	coachini	moderat	49
20	other	b.tech	6.94	67.8	group st	moderat	49.2
22	other	b.tech	3.25	74.8	coachini	easy	49.5
19	other	diploma	0.68	64.3	mixed	moderat	49.9
20	other	b.tech	4.17	47.7	online vi	moderat	49.9

- Total Qualified Students sample

course	exam_sco	count(*)
b.sc	100	2
ba	100	2
b.tech	100	2
bca	100	2
diploma	100	1
b.com	100	3
oba	100	2
diploma	99.3	1
oba	98.5	1
oba	97.9	1
b.tech	97.5	1
oba	97.5	1
oba	97.1	1
ba	96.4	1
b.sc	96.3	1
b.tech	96.1	1
oba	96	1
b.com	95.6	1
diploma	95.5	1
b.sc	95.2	1
ba	93.8	1
bca	93.8	1
diploma	93.1	1
ba	92.2	1

- **Total Unqualified Students sample**

course	exam_score	count(*)
bca	19.599	1
ba	19.599	1
b.tech	19.599	1
diploma	19.599	1
b.sc	19.599	1
bca	19.7	1
b.tech	19.8	1
b.tech	21.4	1
b.tech	22.1	1
b.tech	22.9	1
bba	23.5	1
b.tech	23.7	1
diploma	23.9	1
ba	25.5	1
bba	26.4	1
diploma	26.5	1
diploma	26.7	1
b.sc	27	1
b.tech	27.2	1
b.tech	27.6	1
bba	27.7	1
b.tech	28	1
b.com	28.3	1
bca	28.3	1

PYTHON

In Python, We used Rule Based Condition for predicting the Students performance based on their course such as Top Scorer, Average Scorer and Poor Scorer predicted for individual course. First define a function for the row wise prediction for the dataset. Then set if-elif-else conditions for individual course as per the exam score for the prediction.

```
# Let's do the prediction based on Rule-based Condition
def predict_exam_performance(row):
    if row['course'].lower() == 'bca':
        if row['exam_score'] >= 85:
            return "bca Top Scorer"
        elif row['exam_score'] < 85 and row['exam_score'] >= 50:
            return "bca Average Scorer"
        else:
            return "bca Poor Scorer"
    elif row['course'].lower() == 'b.tech':
        if row['exam_score'] >= 85:
            return "b.tech Top Scorer"
        elif row['exam_score'] < 85 and row['exam_score'] >= 50:
            return "b.tech Average Scorer"
        else:
            return "b.tech Poor Scorer"
    elif row['course'].lower() == 'b.com':
        if row['exam_score'] >= 85:
            return "b.com Top Scorer"
        elif row['exam_score'] < 85 and row['exam_score'] >= 50:
            return "b.com Average Scorer"
        else:
            return "b.com Poor Scorer"
    elif row['course'].lower() == 'bba':
        if row['exam_score'] >= 85:
            return "bba Top Scorer"
        elif row['exam_score'] < 85 and row['exam_score'] >= 50:
            return "bba Average Scorer"
        else:
            return "bba Poor Scorer"
    elif row['course'].lower() == 'diploma':
        if row['exam_score'] >= 85:
            return "diploma Top Scorer"
        elif row['exam_score'] < 85 and row['exam_score'] >= 50:
            return "diploma Average Scorer"
        else:
            return "diploma Poor Scorer"
    elif row['course'].lower() == 'b.sc':
        if row['exam_score'] >= 85:
            return "b.sc Top Scorer"
        elif row['exam_score'] < 85 and row['exam_score'] >= 50:
            return "b.sc Average Scorer"
        else:
            return "b.sc Poor Scorer"
    elif row['course'].lower() == 'ba':
        if row['exam_score'] >= 85:
            return "ba Top Scorer"
        elif row['exam_score'] < 85 and row['exam_score'] >= 50:
            return "ba Average Scorer"
        else:
            return "ba Poor Scorer"
    else:
        return "Other Course"
```

- Then Create a new column with name “prediction”, then applying “.apply()” function and call out the defined function ‘predict_exam_performance’ for prediction through row by row.

```
# Create new column prediction to apply the defined function into the dataset
data['Prediction'] = data.apply(predict_exam_performance, axis = 1)
```

```
# First 50 rows of the predicted Dataset
data.head(50)
```

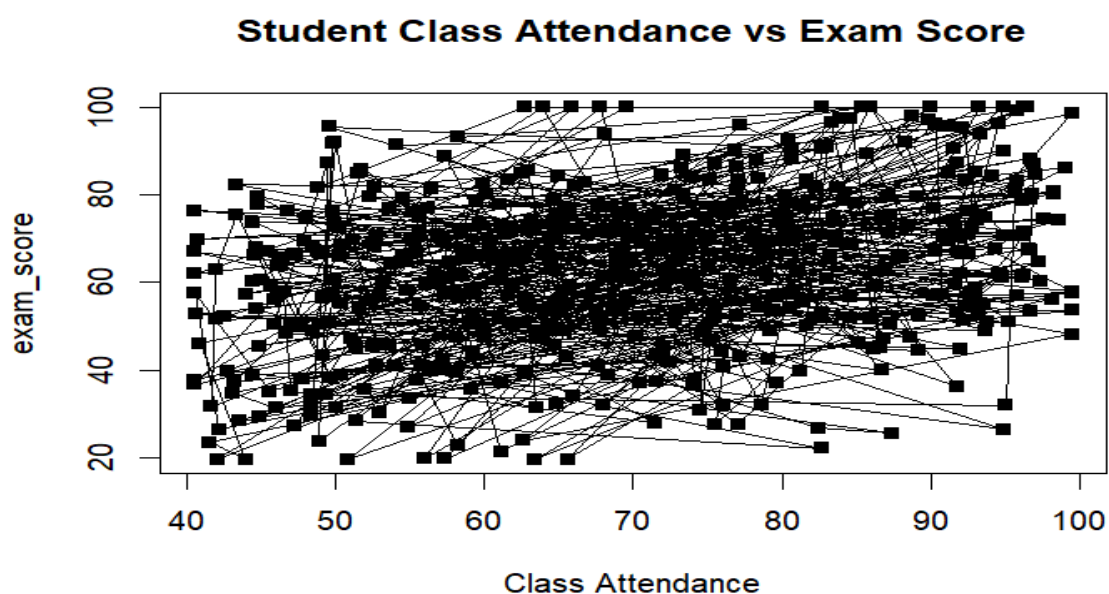
id	gender	course	study_hours	class_attendance	internet_access	sleep_hours	sleep_quality	study_method	facility_rating	exam_difficulty	exam_score	Prediction
17	male	diploma	2.78	92.9	yes	7.4	poor	coaching	low	hard	58.9	diploma Average Scorer
23	other	bca	3.37	64.8	yes	4.6	average	online videos	medium	moderate	54.8	bca Average Scorer
22	male	b.sc	7.88	76.8	yes	8.5	poor	coaching	high	moderate	90.3	b.sc Top Scorer
20	other	diploma	0.67	48.4	yes	5.8	average	online videos	low	moderate	29.7	diploma Poor Scorer
												diploma

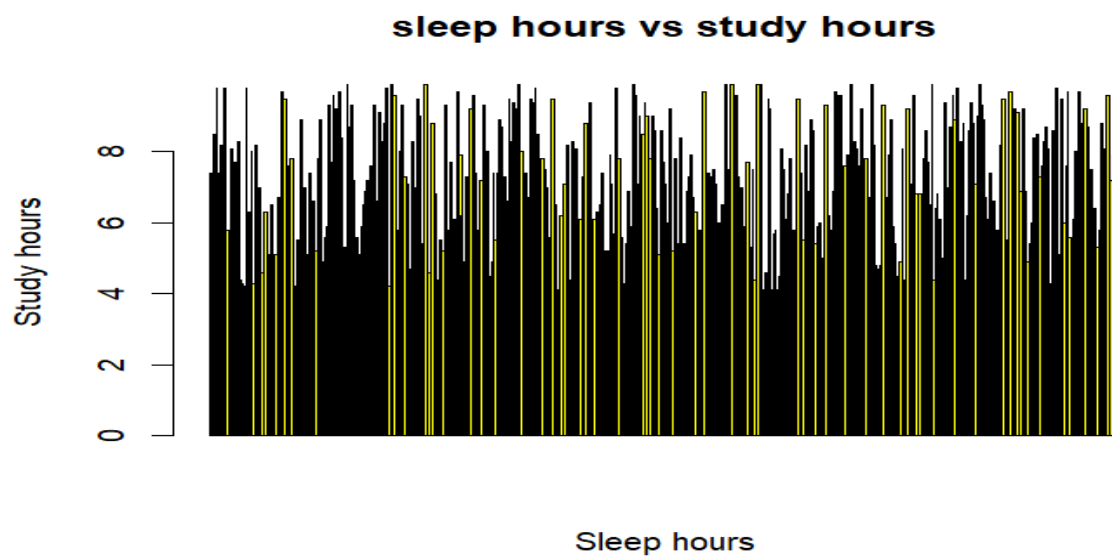
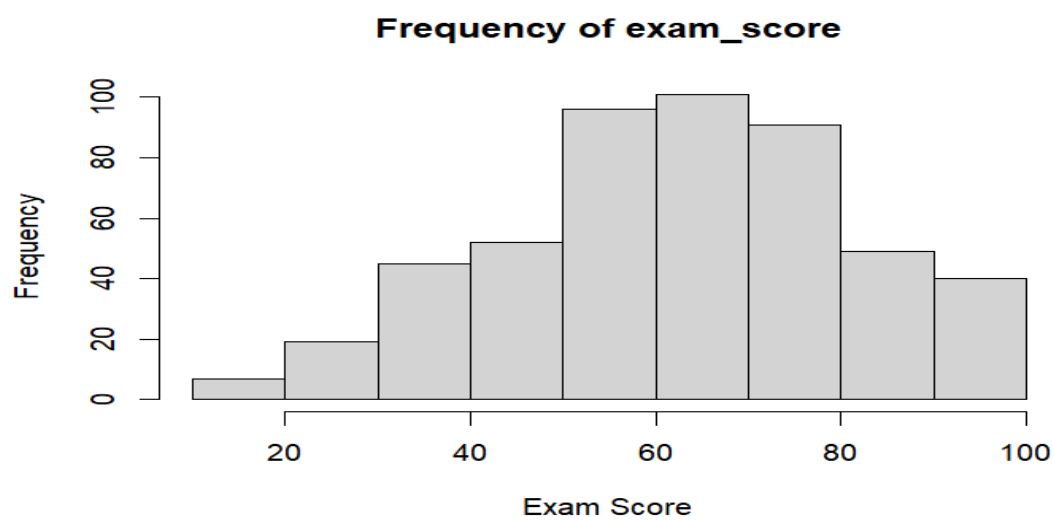
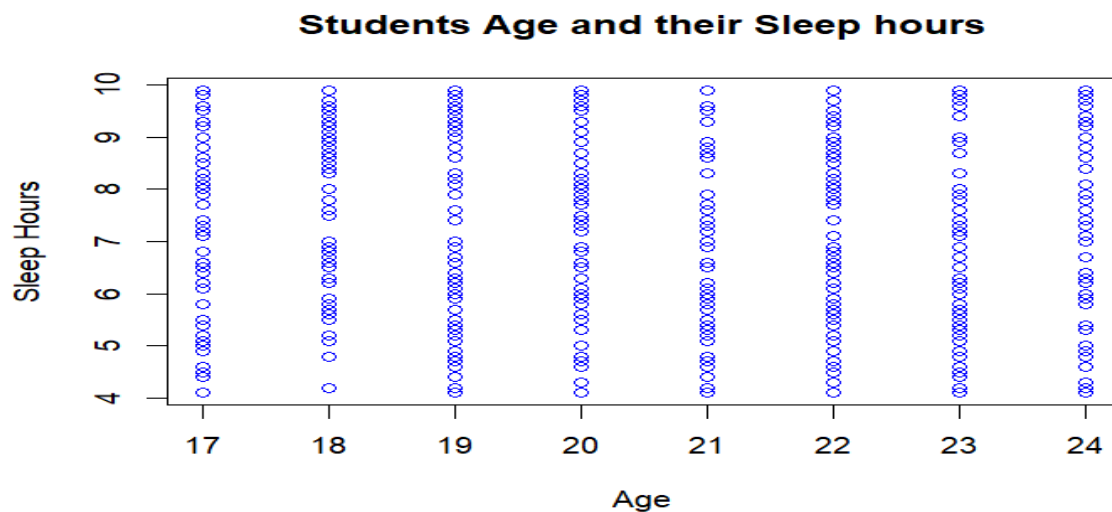
```
# Lets save this Exam_score with prediction dataset to csv
data.to_csv("student_performance_with_prediction.csv", index=False)
```

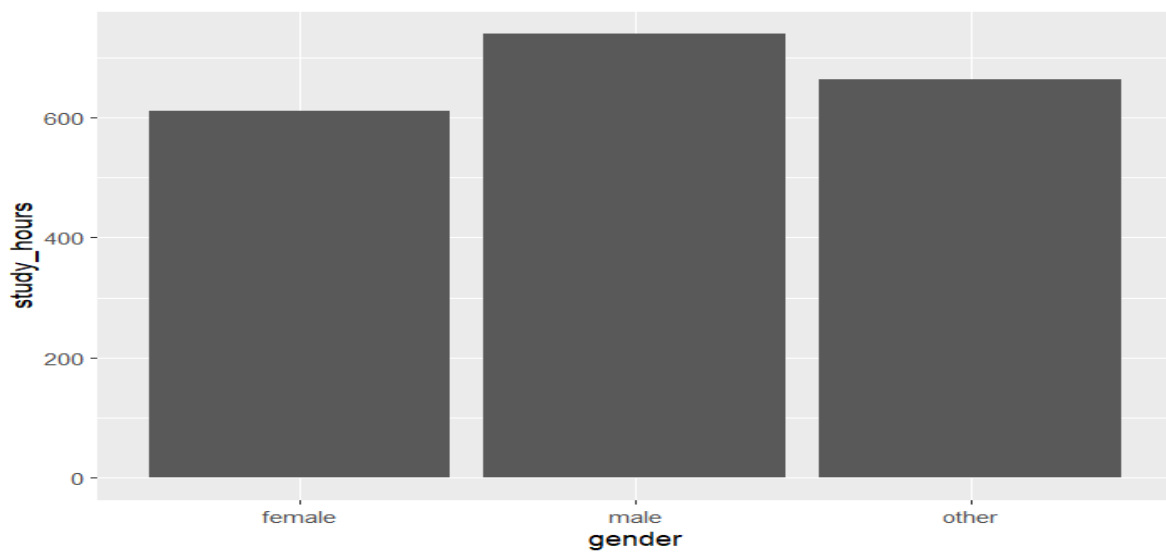
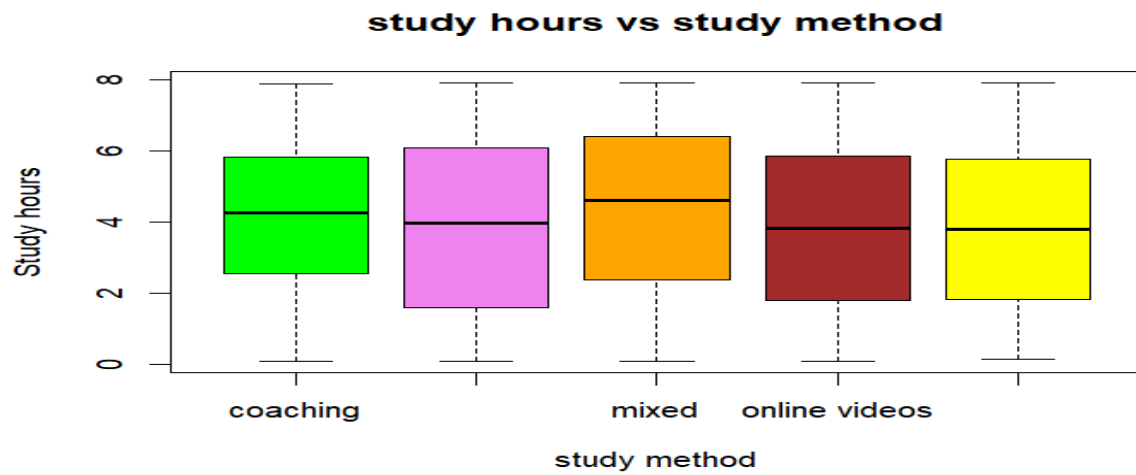
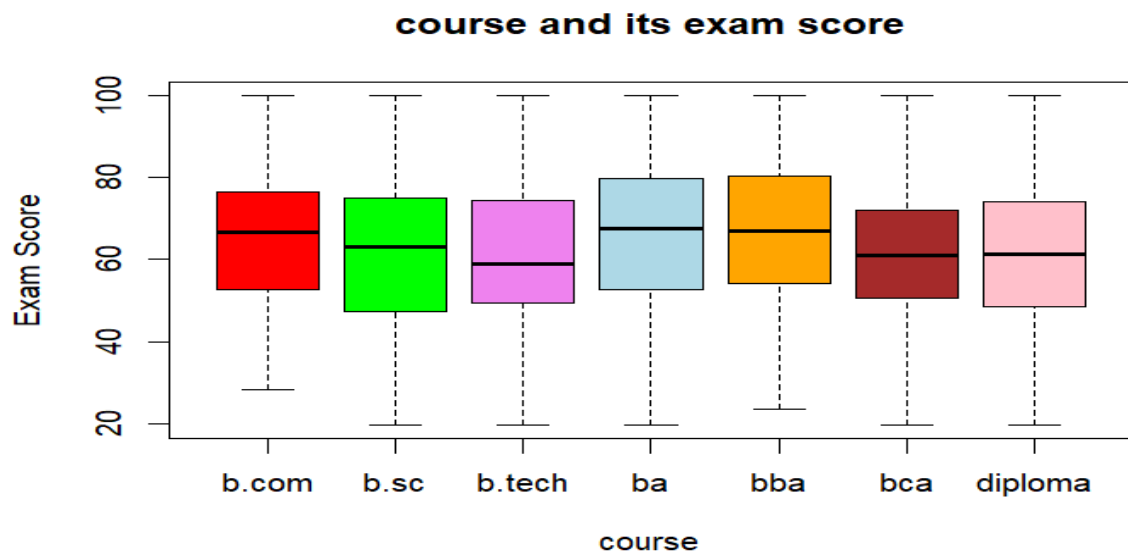
- Lastly with prediction column, Save the dataset for the further Process.

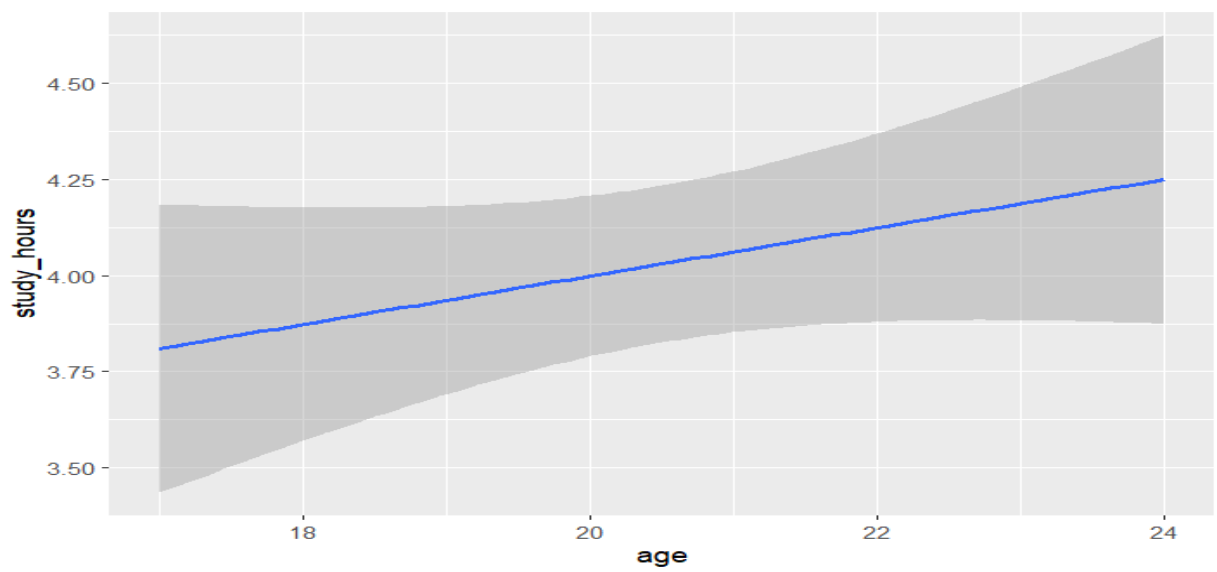
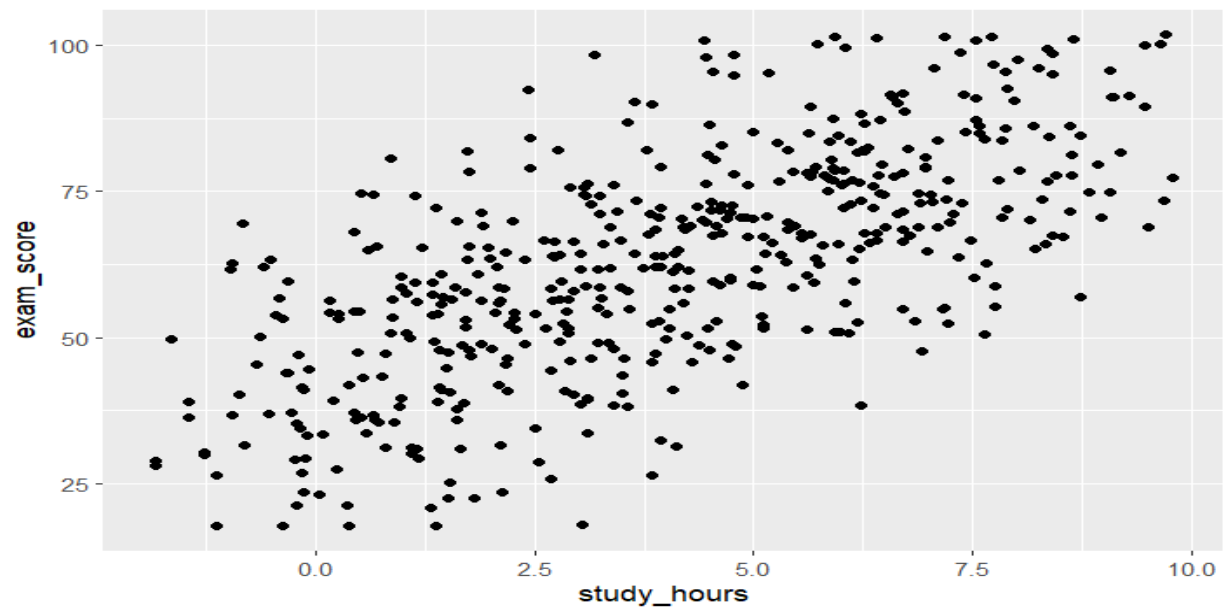
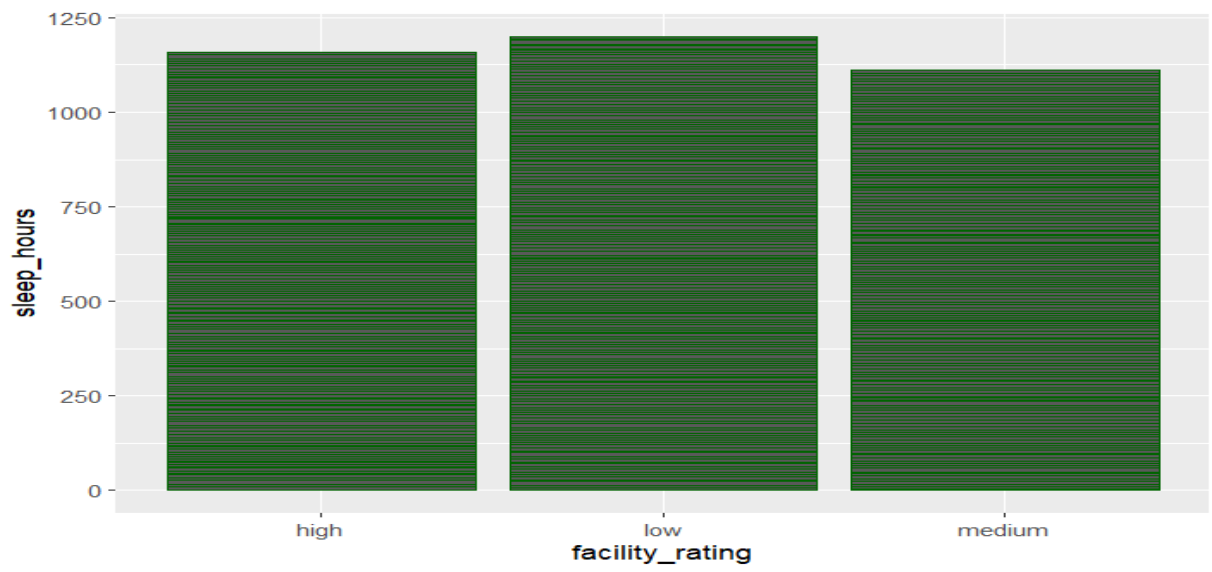
R

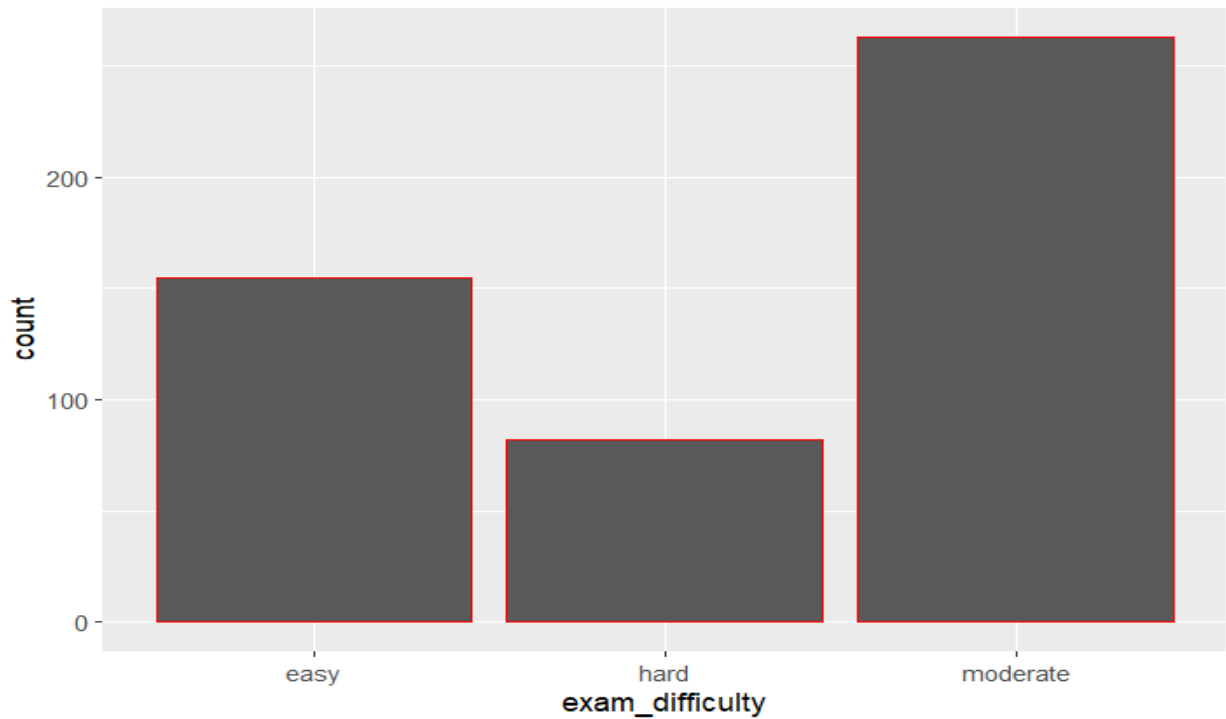
With this predicted dataset, we’ll visualize the data using lineplot, Scatterplot, Barplot, Boxplot, histogram and ggplot(geom_bar, geom_col, geom_jitter).











Conclusion:

Through this Student Performance Analysis, we Successfully Managed the student database as per our requirements. We did simple effective prediction using Rule- Based Prediction, without using any Machine Learning Algorithm and Lastly Visualized with Attractive Plots that helps to understand the data distribution. For the 100 Scorers compare to Male, Female Scored better with qualifying most moderate level exam. For 95-99 scorers Male performed better. Other gender scored poor with Not qualifying Easy level exam compare to Male and Female.